Meet-EU · BRCA1 variant project · Kickoff

BRCA1: from DNA repair to cancer, and the variants we can’t yet understand

Marian Novotný · Charles University, Prague · 16 October 2026

One gene, three numbers

1994

BRCA1 cloned, the first major breast cancer susceptibility gene. BRCA2 followed in 1995.

72%

Breast cancer risk by age 80 in carriers of a pathogenic germline variant.

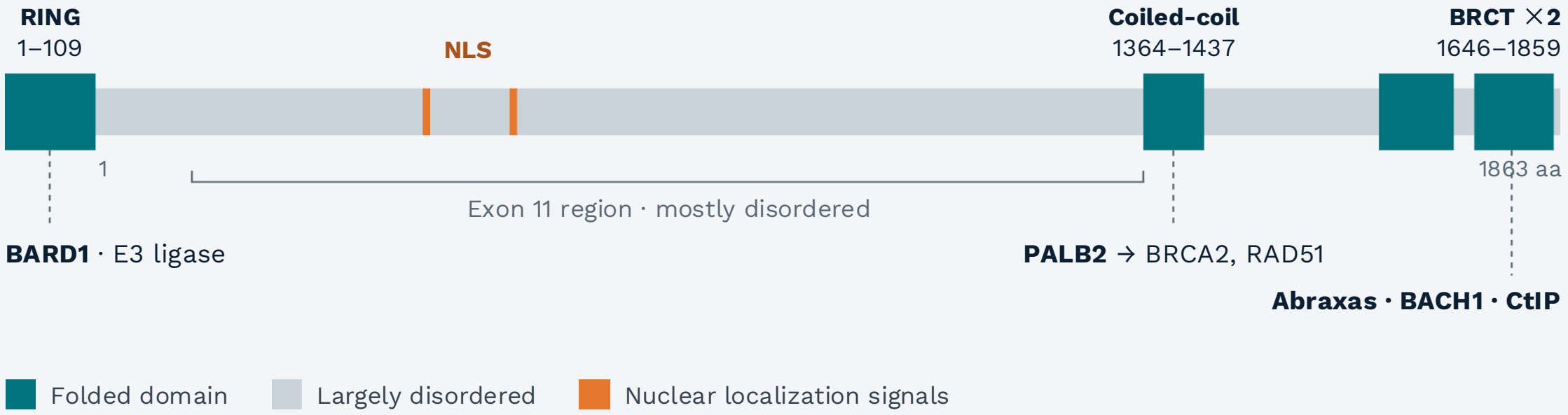
44%

Ovarian cancer risk by age 80 in the same carriers.

BRCA1 and BRCA2 share a name, not an ancestry: they are unrelated proteins working in the same DNA repair pathway.

Risks: Kuchenbaecker et al., JAMA 2017

BReast CAncer gene 1



UniProt P38398 · drawn to scale

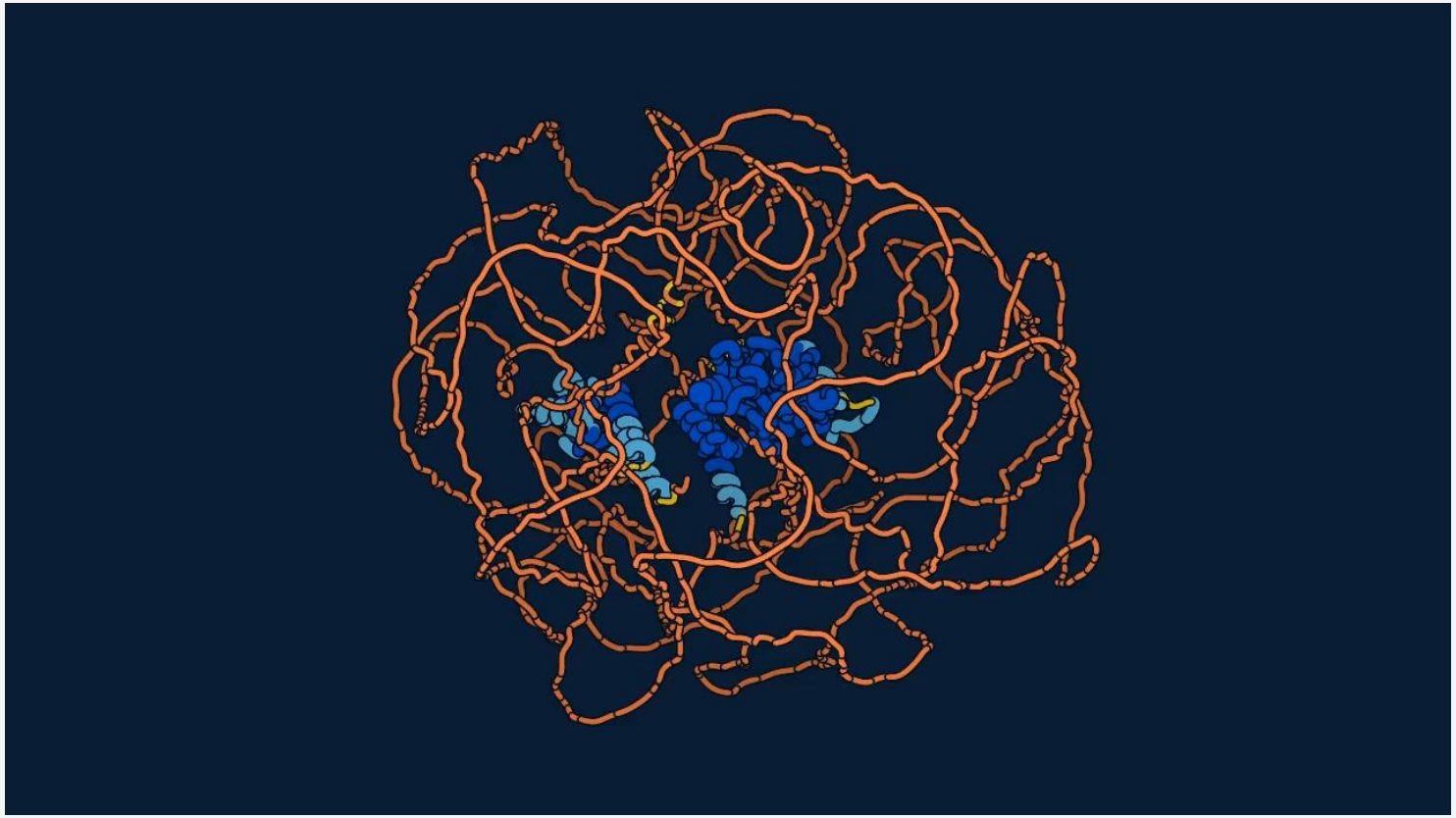
1,863 residues, and most of them never fold

80%

of residues predicted with very low confidence (pLDDT under 50)

Mean pLDDT · BRCT 91 · RING 81 · coiled-coil 54 · middle 31

- Very high, over 90
- Confident, 70–90
- Low, 50–70
- Very low, under 50



AlphaFold DB, AF-P38398-F1 (model v6) · rendered as a smoothed Cα trace

Repairing double-strand breaks without errors

01
Break
Both DNA strands cut

02
BRCA1 decides
Outcompetes 53BP1, chooses HR

03
Resection
CtIP and MRN expose single strands

04
Handoff
PALB2 → BRCA2 loads RAD51

05
Copy
Sister chromatid as template

> **Without BRCA1:** 53BP1 wins, breaks go to error-prone end joining, and the genome slowly falls apart.

Animation: WEHI, “DNA break repair by homologous recombination” (Drew Berry, 2024)

A hub with many side jobs

E3 ubiquitin ligase

With BARD1, ubiquitinates histone H2A (structure 7JZV)

Checkpoints

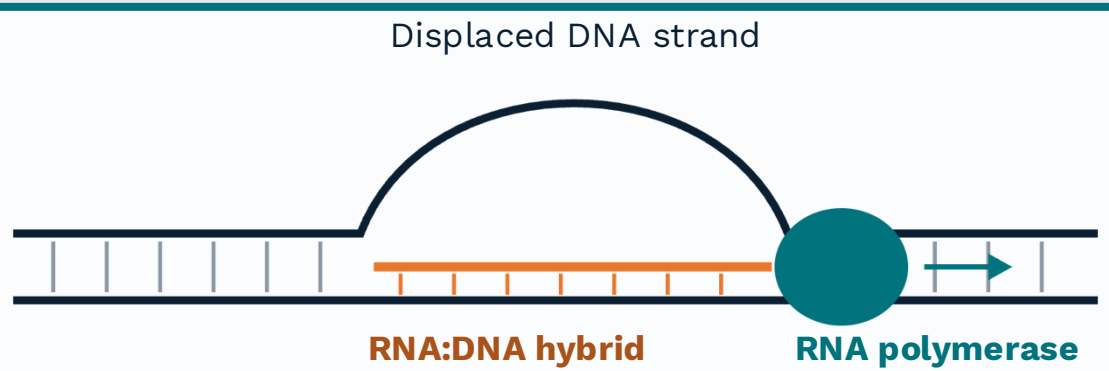
Arrests the cycle in S and G2/M after damage

Replication forks

Protects stalled forks from MRE11 degradation

Transcription and R-loops →

Helps remove RNA:DNA hybrids before they break DNA



Estrogen-driven transcription makes R-loops more frequent. BRCA1 recruits senataxin (SETX) to resolve them; left in place, they collide with replication and become breaks.

R-loops and BRCA1: Hatchi et al., Mol Cell 2015

From lost repair to cancer: one inherited hit is not enough

1

Inherited hit

One mutant allele in every cell of the body

2

Second hit

The normal allele is lost, usually by LOH

3

p53 lost

The cell survives without HR instead of dying

4

Genomic chaos

HRD scars (signature SBS3) and a tumor

> **A silent route:** promoter methylation switches BRCA1 off without any mutation, with the same HRD phenotype.

Share of tumors with a germline BRCA1 mutation

Tumor type	Germline BRCA1	Worth knowing
High-grade serous ovarian	8–12%	Plus ~10–15% promoter methylation
Triple-negative breast	10–15%	20–30% in women under ~40
Breast, unselected	1–3%	BRCA1 and BRCA2 roughly equal
Male breast	1–4%	BRCA2 dominates (~10%)
Pancreatic	0.5–1%	BRCA2 more common (~2–4%)
Prostate, metastatic	0.5–1%	BRCA2 more common (~5%)

Approximate ranges; they vary with population and patient selection.

BRCA1 works in every cell. Why breast and ovary, not lung or kidney?

A **Hormones drive damage**

Cyclic estrogen and progesterone push proliferation; estrogen-driven transcription creates R-loops.

B **A vulnerable cell of origin**

Luminal progenitors in the breast; secretory cells of the fallopian tube, exposed at every ovulation.

C **Losing BRCA1 usually kills**

Most cells die without HR. A tumor needs a tissue where the cell survives it, with p53 gone too.

D **Other tissues, other engines**

Lung runs on smoking and KRAS or EGFR; kidney on VHL loss. HR loss gives them little advantage.

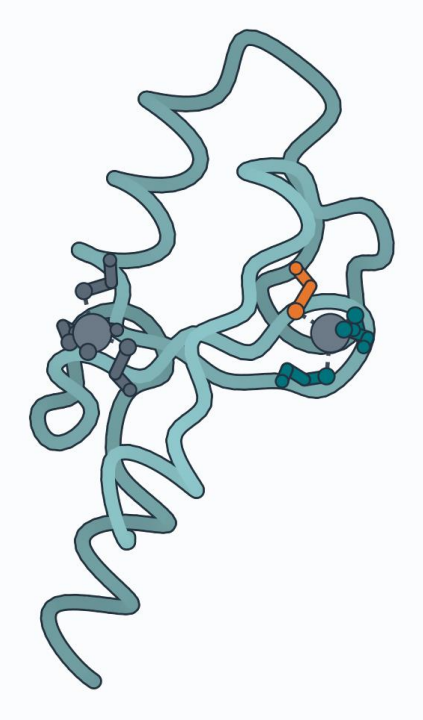
Still an open question. Same pattern elsewhere: RB1 → retina, VHL → kidney.

Founder mutations: one ancestor, many carriers

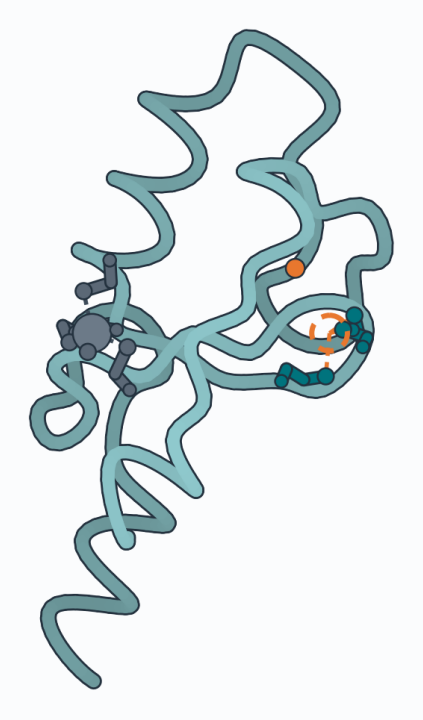
Population	Variant	Why it is interesting
Ashkenazi Jewish	c.68_69delAG (185delAG), c.5266dupC	~2.5% carry one of three BRCA1/2 founders
Central Europe, Czechia	c.5266dupC, c.181T>G (C61G), c.3700_3704del	C61G: missense that breaks the RING zinc site
Spain (Galicia)	c.211A>G (R71G)	Looks like missense, acts through splicing
Greece	c.5212G>A (G1738R)	Missense that destabilizes the BRCT fold
Quebec	c.4327C>T (R1443X)	From a few 17th-century settlers
Mexico	Deletion of exons 9–12	Missed by sequencing; needs MLPA

Check individual variants in ClinVar or BRCA Exchange.

Pathogenic mutation - One atom, one zinc, one lost protein: C61G



Wild type · Cys61



C61G · glycine, no side chain

Zinc ion Zinc 2 ligands Residue 61

1 **Cys61's sulfur holds zinc 2**, together with Cys39, His41 and Cys64.

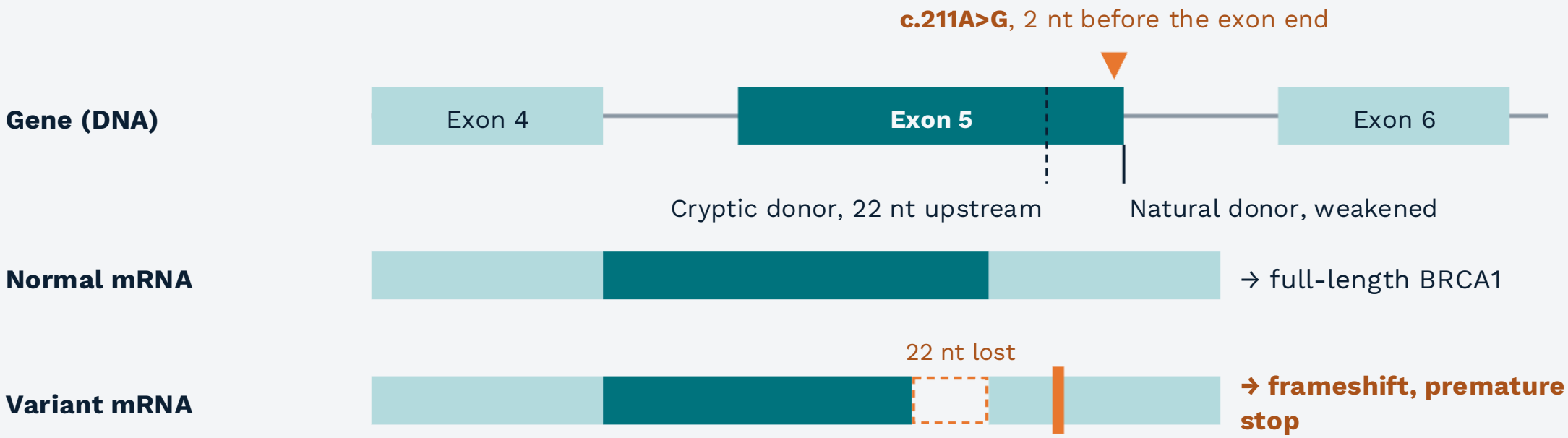
2 **Glycine has no side chain**, so the site loses a ligand and cannot hold its zinc.

3 **The RING loses its fold:** BARD1 binding and E3 ligase activity break down.

4 **Pathogenic**, and a founder variant in Czechia and Poland.

AlphaFold AF-P38398-F1, residues 14–82 · zinc placed between its ligands (no metals in the model) · mutant panel is a schematic

A “missense” variant that breaks splicing: c.211A>G



> Read as protein, it is a single Arg71Gly change. Read as RNA, it destroys the reading frame. That is why the project has an RNA-level mechanism class.

Spanish founder variant; Vega et al. 2001 · legacy exon numbering

ClinVar: where variant verdicts are recorded

Five classes, from one or many labs

A public NCBI archive where diagnostic labs and expert panels submit their interpretation of a variant.

Pathogenic	Likely pathogenic	Uncertain (VUS)	Likely benign	Benign

When submitters disagree, the record reads **“conflicting classifications”**.

Review status: how much to trust it

★★★★ Practice guideline

★★★ Expert panel (for BRCA1: ENIGMA)

★★ Several labs, no conflict

★ One lab, or labs in conflict

No stars: no criteria given

6,463
BRCA1 records in our frozen pull

4,606
to predict: 1,932 VUS, 2,674 conflicting

696
reference labels: 232 P/LP, 464 B/LB

Thousands of variants, no verdict

4,606

BRCA1 missense variants in ClinVar without a verdict: 1,932 uncertain and 2,674 conflicting

Saturation genome editing

- ~3,900 SNVs tested in the native gene (HAP1 cells)
- Covers the RING and BRCT exons
- ~97% agreement with expert classifications
- Showed many variants act through splicing

The disordered middle, where most VUS sit, is untested.

ClinVar snapshot of 5 October 2026 · Findlay et al., Nature 2018

02

The project

Beyond the VUS: predicting the effect, and the mechanism, of BRCA1 missense variants

Meet-EU · Prague · Warsaw · Copenhagen · Paris · Milan · Heidelberg

One gene you can hold in your head

01

Real stakes

A pathogenic call changes surgery, screening, PARP inhibitor eligibility and family testing.

02

An honest test

Two folded islands in a disordered sea. You will run into the limits of the field, which is the point.

03

Prospective, like CAGI

Predictions are frozen on a fixed date. Months later, we see how ClinVar reclassified them.



The “Angelina effect”

In 2013 Angelina Jolie, a BRCA1 carrier, wrote publicly about her preventive double mastectomy. Requests for BRCA testing rose sharply in the following months.

Jolie, “My Medical Choice”, New York Times, 14 May 2013

CAGI: predict first, learn the answer later

01

Hidden data

A lab holds unpublished experimental or clinical results.

02

Blind predictions

Teams worldwide submit predictions without seeing them.

03

Independent assessors

Scored by people who did not build any of the methods.

04

Open discussion

A joint meeting and papers on what worked and why.

CAGI7 “Annotate All Missense”

Score every possible human missense variant: 0–1, an SD, and a damaging, tolerated or unknown call. Scored against annotations that appear **after** submission.

Our version

Same file format, one protein. Frozen on 26 January 2027, checked against ClinVar 6–12 months later.

Is it damaging? And if so, why?

1 · Effect

A probability with an uncertainty, not a label.

2 · Mechanism

As specific as your evidence allows, and no more.

Binding disruption · parent class

Zinc site	BARD1 interface
BRCT phosphopeptide pocket	PALB2 coiled-coil

Fold destabilization	Phosphosite loss
Disrupts splicing	Damaging, mechanism unknown
No effect	

Resolution is rewarded, guessing is not: “damaging, mechanism unknown” beats an invented story.

Full taxonomy: project plan, section 3

A possible skeleton of the project

1	2	3	4	5	6
Curate	**Features**	**Baselines**	**Model**	**Validate**	**Predict**
Frozen dataset, MaveDB, gnomAD	Evolution, structure, interfaces, RNA	Worth reproducing two published predictors	Your choice; logistic regression is fine	Your own train, validation and test split	All 4,606 variants, mechanism for each damaging call

> Use it, adapt it or replace it. What you must declare is how you split your data, before you start modelling.

Details: project plan, section 4

Reasoning with data, not calling an API

Combine evidence

At least two independent kinds, never one tool’s output.

State uncertainty

A well-placed “unknown” beats a confident mistake.

Try baselines

Compare with two published predictors on the same data.

Name the biology

A mechanism points to a residue, structure, motif or assay.

Mind circularity

Say which features could have seen your evaluation labels.

Declare AI use

Expected; say what you used and for what.

Details: project plan, sections 5 and 8

Not a competition

You hand in

- Prediction file for all 4,606 variants (CAGI style)
- 10 mechanism cards
- Split policy file, written before modelling
- Methods description, 3–5 pages
- Code repository in the course GitHub organization:
  https://github.com/meet-eu-26-27
- Short oral summary, then a poster session

What we look at

- Discrimination on the 696 reference variants
- Coverage and honest abstention
- Mechanism plausibility
- Your split design, and its optimism gap

> Nothing is withheld, so we compare how you validated, not who scored highest. Months later, we check what ClinVar did with your variants.

Details: project plan, sections 7 and 9 · grades are set locally

Schedule and practicalities

16 Oct 2026

Kickoff

Today, 14:00 · Zoom. Teams formed, data distributed

4 Dec 2026

Mid-term

14:00 · Zoom. Baselines and split design, peer review

26 Jan 2027

Freeze

Everything handed in; nothing changes after

29 Jan 2027

Final workshop

Full day, in person, Warsaw. Talks and posters

Webpage: https://meet-eu-26-27.github.io/

Team registration:
https://docs.google.com/spreadsheets/d/1YvdEg2LuC1FmioXVU7MOplCH4hLe4XfTSosGODr3Ip8/edit?gid=0#gid=0